# OOD Quantification → Category-Shift Adaptation

Feature- & sample-level weights with a soft$^\gamma$ cold-start gate

AS-DTA / CFPerm

**Abstract**

**Message.** Category shift is an OOD event on the live stream. We *quantify* the footprint once (RF-domain VIMP + membership), then *adapt* a CE head with two cheap weights via `partial_fit`—no new backbone, tiny compute. v3 uses a **soft$^\gamma$ cold-start gate**

$$\text{decay} = \big((1 - n_{\text{idx}}/N_{\text{cap}})_+\big)^\gamma, \quad N_{\text{cap}} = 200,\ \gamma = 0.5 \text{ (linger)}.$$

$\gamma$<1 keeps empowerment longer than linear taper; $\gamma$>1 cuts faster. On text multiclass (FULL LIVE holdout, mean±SEM over boards), early overall-acc lift is clear; soft$^{0.5}$ is the best gated default.

## 1 OOD quantification → category-shift adapt

**Problem.** REF covers known categories; LIVE mixes known support with novel labels. Question: *where is the OOD mass, and how do we give it CE budget cheaply?*

**Quantify.** Score $W \in \{0, 1\}$ (REF vs LIVE) with a shallow RF. Importances $v \in \Delta^{p-1}$ are the OOD coordinates. On top-$K$ of $v$, $z$-distance to REF yields membership $m_i \in [0, 1]$ ("new-category-like").

**Adapt.**

- **Feature-level:** stretch OOD axes $x_{ij} \leftarrow x_{ij}(1 + \alpha_{\text{eff}}\tilde{v}_j)$.
- **Sample-level:** membership excess $\times \text{decay}_i$ as `sample_weight` in `partial_fit`.

One RF pass + a few linear CE epochs. No EWMA, no architecture change.

**Soft$^\gamma$ gate (v3).**

$$\text{decay}_i = \big((1 - n_{\text{idx}}(i)/200)_+\big)^\gamma \quad \text{on novel rows.} \tag{1}$$

Default $\gamma$=0.5 *lingers*: scarce regime stays empowered longer than the linear ($\gamma$=1) schedule from v2. Fast taper $\gamma$=2 and always-on are ablations.

## 2 Learning curves (FULL LIVE holdout)

KPI = overall accuracy / mean novel OVA-AUC on a fixed full LIVE holdout. Shaded bands = SEM across datasets. Dashed line = $N_{\text{cap}}$.

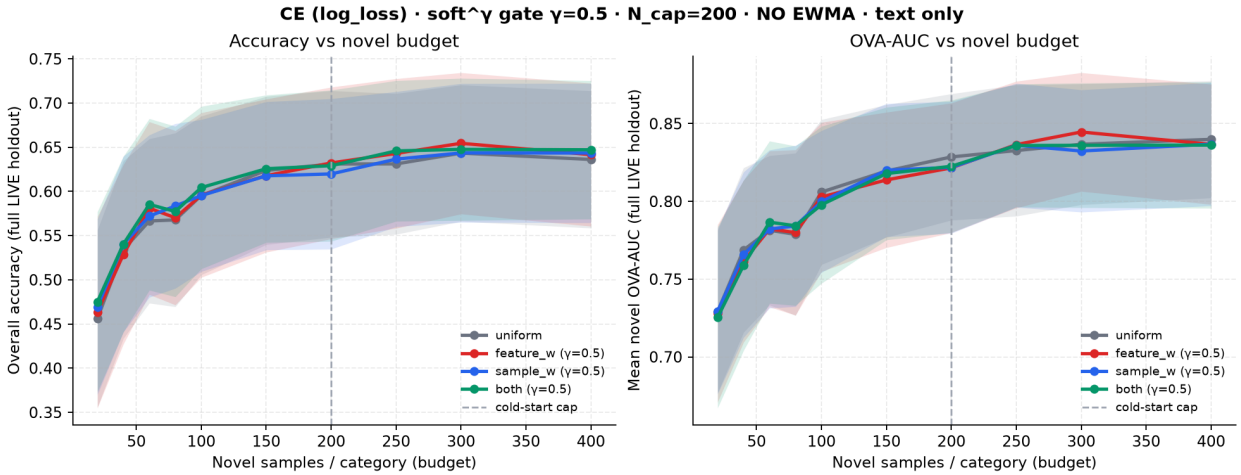



Figure 1: Default soft$^{\gamma=0.5}$ schemes vs uniform (mean±SEM). Early: weighted schemes lead. Past 200: soft$^{0.5}$ stays above uniform on overall acc.

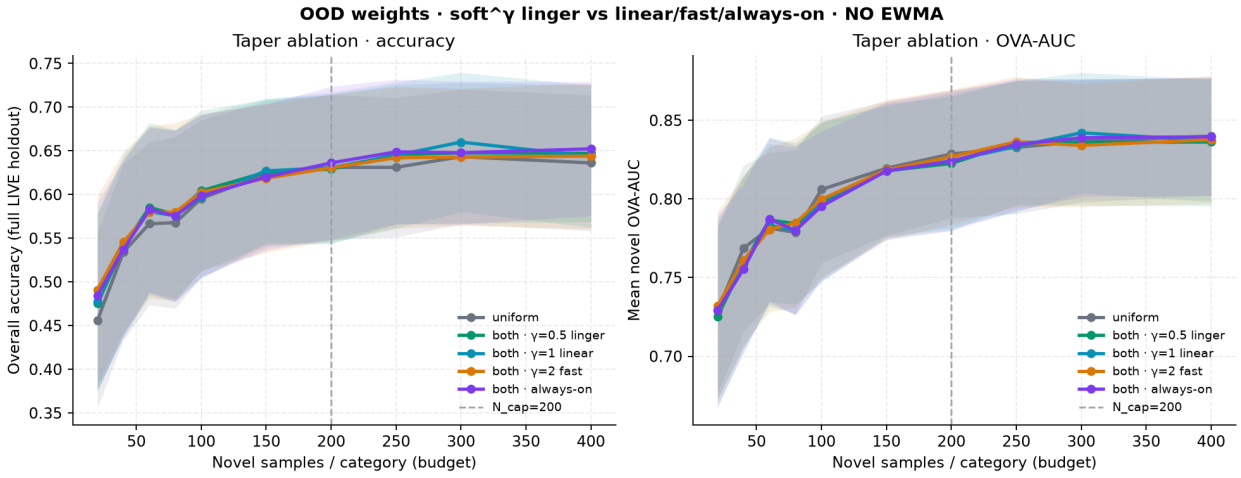



Figure 2: Taper ablation on `both`: $\gamma$=0.5 linger (default), $\gamma$=1 linear, $\gamma$=2 fast, always-on. Linger is the best gated choice; always-on can keep a small late edge but never stops boosting warm categories.

**Reading the figures.**

- **Empowerment.** At $b$=20/cat, soft$^{0.5}$ `both`: 0.456 → **0.475** ($\Delta$+0.019).
- **Linger wins among gates.** Endpoint mean $\Delta$acc vs uniform: soft$^{0.5}$ `both` +**0.011**, $\gamma$=1 +0.007, $\gamma$=2 +0.008.
- **OVA-AUC flat.** Operational gain is overall acc under CE.

## 3 Endpoint + ops rule

| | uniform | feature $\gamma$=0.5 | sample $\gamma$=0.5 | both $\gamma$=0.5 |
|---|---|---|---|---|
| mean overall acc | 0.636 | 0.642 | 0.643 | **0.647** |
| mean $\Delta$acc vs U | — | +0.005 | +0.007 | +**0.011** |

**Ops rule.**

1. Scarce new category ($n_c \ll 200$): use `both` with soft$^{\gamma=0.5}$.
2. Token-sparse boards (few dims carry the shift): feature-level alone is enough.
3. Past $N_{\text{cap}}$: decay has tapered—no EWMA, no manual schedule; just stop over-weighting.

## 4 Takeaway

1. **Quantify OOD, then adapt** with feature + sample weights.
2. **Ship with `partial_fit`** — tiny compute.
3. **Empower with soft$^{\gamma=0.5}$** — linger through the scarce window, taper by 200 samples/category.